Marine Capture and Household Fish Consumption in North Maluku, 2020

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2.1.1 Species Groups and Landed Volumes

Table 1. The ten species groups, their names in SUSENAS and in English, and their marine capture, North Maluku, 2020.

Species Group	Indonesian Names	English Names	KKP Taxa	Marine Capture (t)	Share of Marine Capture (%)
Thunnini	tongkol, tuna, cakalang	tunas (skipjack, yellowfin, kawakawa, frigate tuna)	7	114,239	35.7
Carangidae	selar, bawal	scads and trevallies	14	63,147	19.7
Other Actinopterygii	*ikan lainnya*	other finfish (12 families, groupers the largest)	23	48,804	15.3
Lutjanidae and other kakap	*kakap*	snappers, with sweetlips, barramundi, and tripletail landed as kakap	8	26,417	8.3
Rastrelliger	*kembung*	Indian mackerels	2	21,108	6.6
Caesionidae	*ekor kuning*	fusiliers	3	15,834	4.9
Engraulidae	*teri*	anchovies	3	11,364	3.6
Cephalopoda	*cumi-cumi*	squid	1	7,705	2.4
Siganidae	*baronang*	rabbitfishes	4	6,079	1.9
Scomberomorus	*tenggiri*	Spanish mackerels	3	5,229	1.6
			68	**319,925**	**100.0**

Findings

- Tuna (Thunnini) landed 114,239 tonnes in 2020, 36% of the marine catch of the province, followed by scads and trevallies (Carangidae) with 20% and other finfish (other Actinopterygii) with 15%. Together these three groups make up 71% of the 319,925 tonnes landed.
- Thunnini reach this volume with only 7 KKP taxa, whereas other Actinopterygii pool 23 taxa for less than half the tonnage, so a few tuna species carry the fishery.

2.1.2 Concentration of Marine Capture

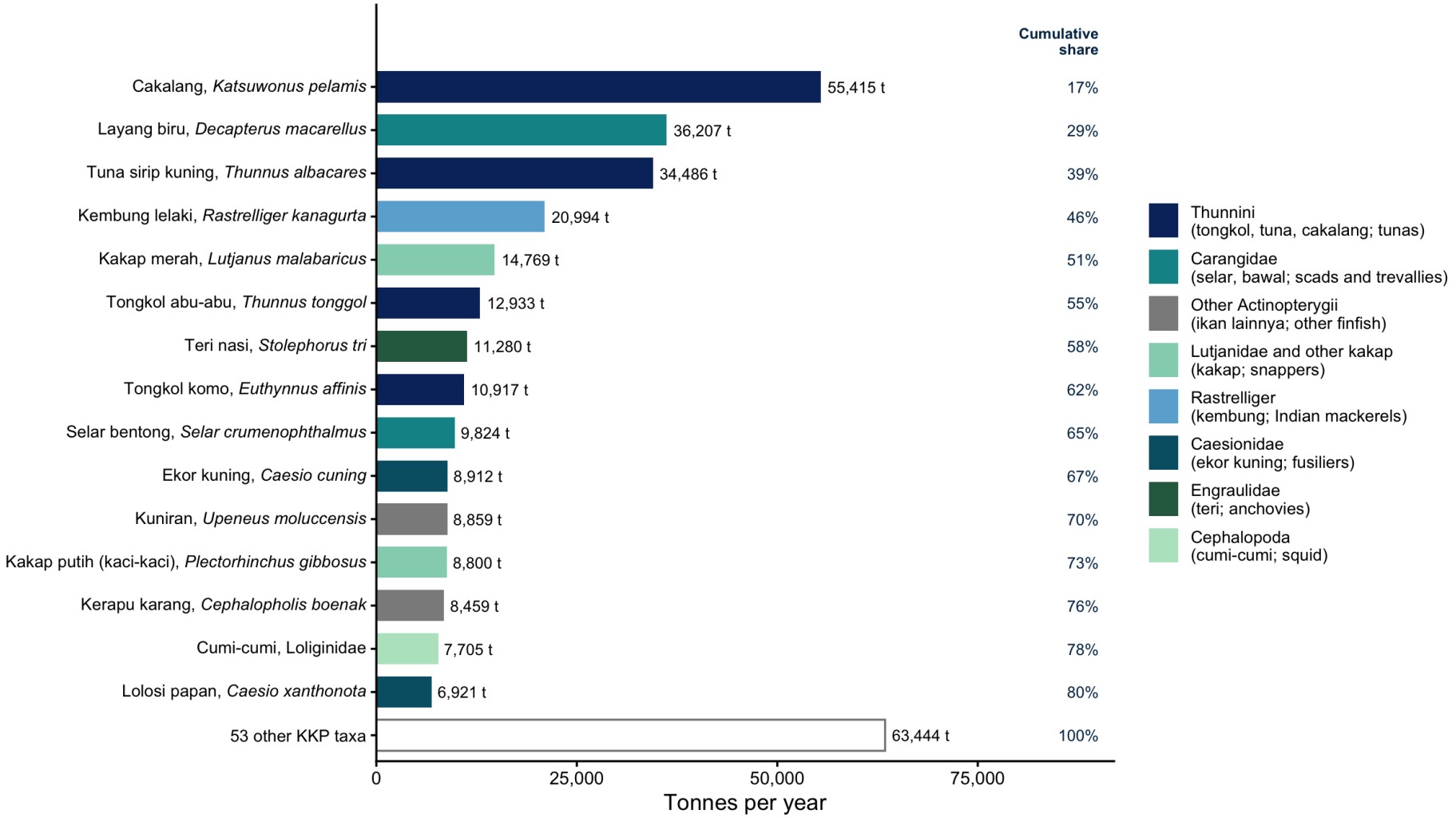


Figure 1. The fifteen largest KKP taxa of marine capture, North Maluku, 2020. Each label gives the market name and the scientific name; bars are coloured by species group, the right-hand column gives the cumulative share, the white bar holds the remaining 53 taxa, and a bracket gives the KKP fish group where two taxa share a name.

Findings

- Cakalang (skipjack tuna), layang biru (mackerel scad), tuna sirip kuning (yellowfin tuna), kembung lelaki (Indian mackerel), and kakap merah (red snapper) make up half of the marine catch, and the fifteen taxa in the figure make up 80%, which leaves 20% for the other 53 taxa.
- Cakalang alone lands 55,415 tonnes, 17% of the total and more than the 51 smallest taxa together, so the catch of North Maluku rests on one tuna species above all.

2.1.3 Marine Capture Across Districts

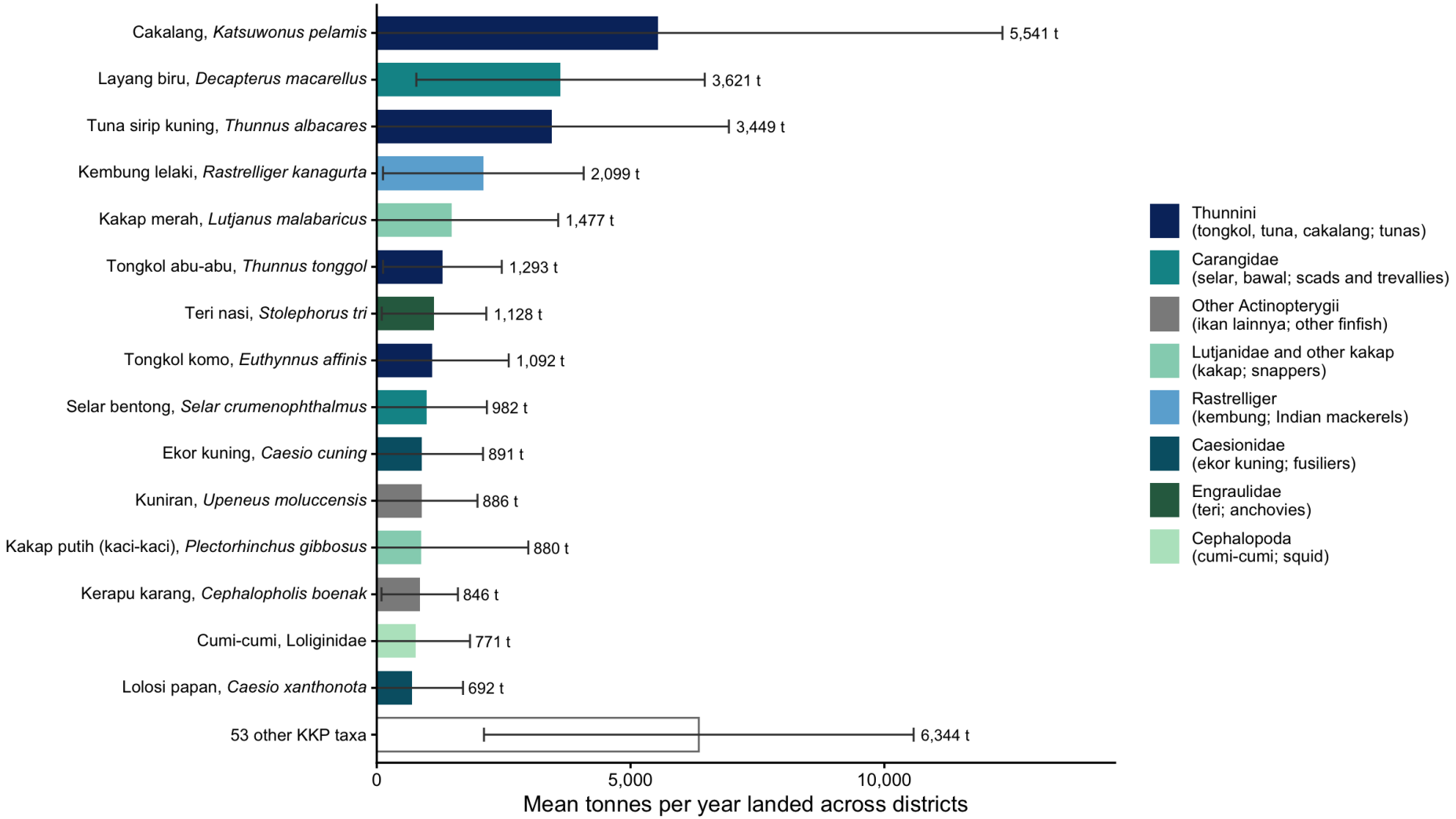


Figure 2. Mean and standard deviation across the ten districts of the landings of the fifteen largest KKP taxa, North Maluku, 2020. The bar is the mean of the ten district landings, the whisker spans one standard deviation on each side of the mean (cut at 0), the label gives the mean, and the white bar holds the 53 other taxa together; order and colours as in Figure 1.

Findings

- The landings of most large taxa are concentrated in a few districts. Ternate lands 24,088 of the 55,415 tonnes of cakalang, Pulau Morotai 11,380 of the 34,486 tonnes of tuna sirip kuning, and Halmahera Selatan 6,844 of the 8,800 tonnes of kakap putih.
- Layang biru is the exception, landed in all ten districts with no district above 8,081 of its 36,207 tonnes, so the ten districts share it more evenly than any other large taxon.

2.1.3 Marine Capture Across Districts

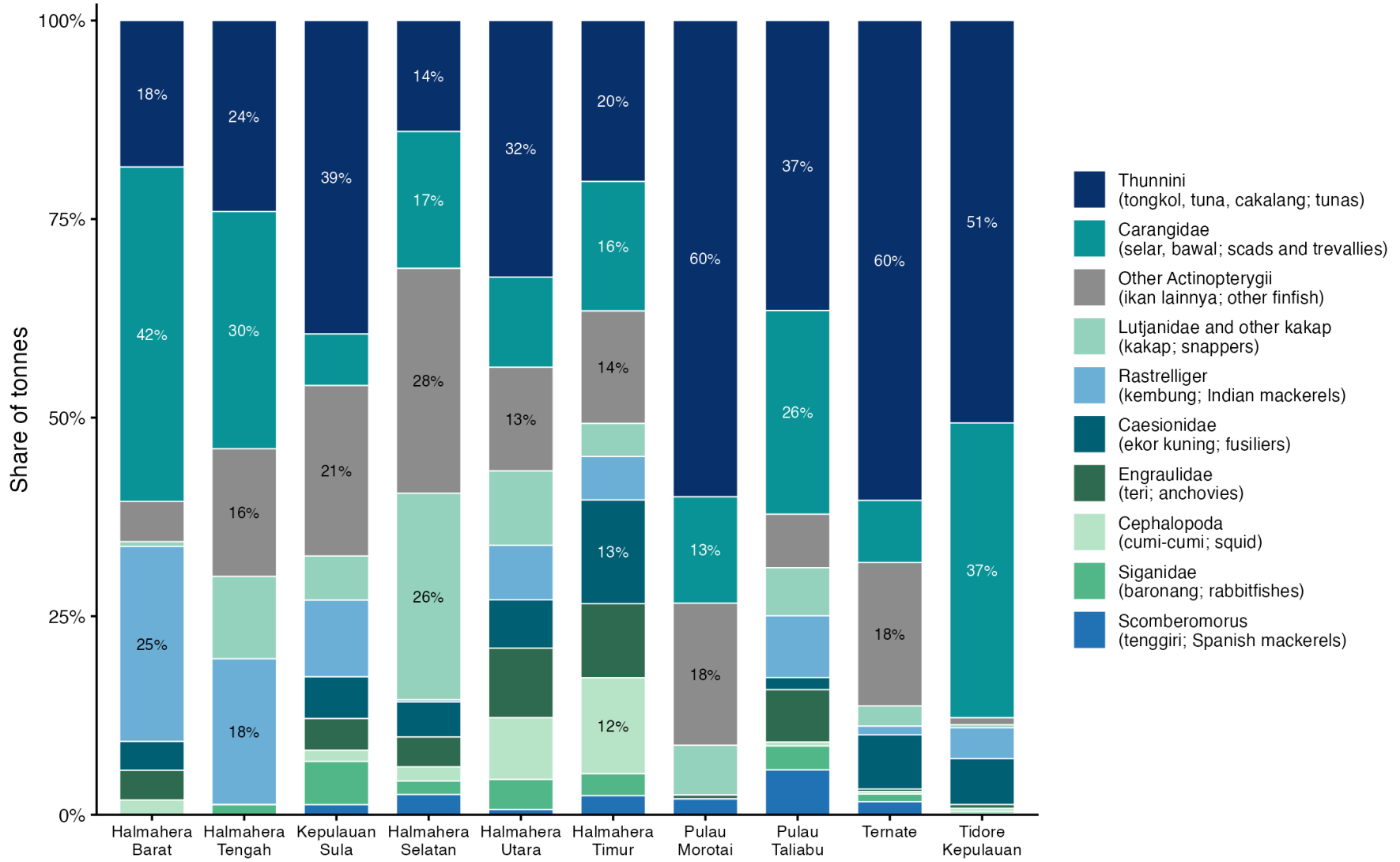


Figure 3. Species composition of marine capture by district, North Maluku, 2020. Shares of at least 12% are labelled.

Findings

- Tuna dominates the landings of Ternate, Pulau Morotai (60% each), and Tidore Kepulauan (51%). In Halmahera Selatan tuna is only 14% of the landings, behind other finfish (28%) and the kakap group (26%), and in Halmahera Barat scads and trevallies (42%) and Indian mackerels (25%) come first.
- Halmahera Timur has the most mixed landings, with no group above 20%, so the species mix of the catch differs from district to district even though tuna leads the province as a whole.

2.1.3 Marine Capture Across Districts

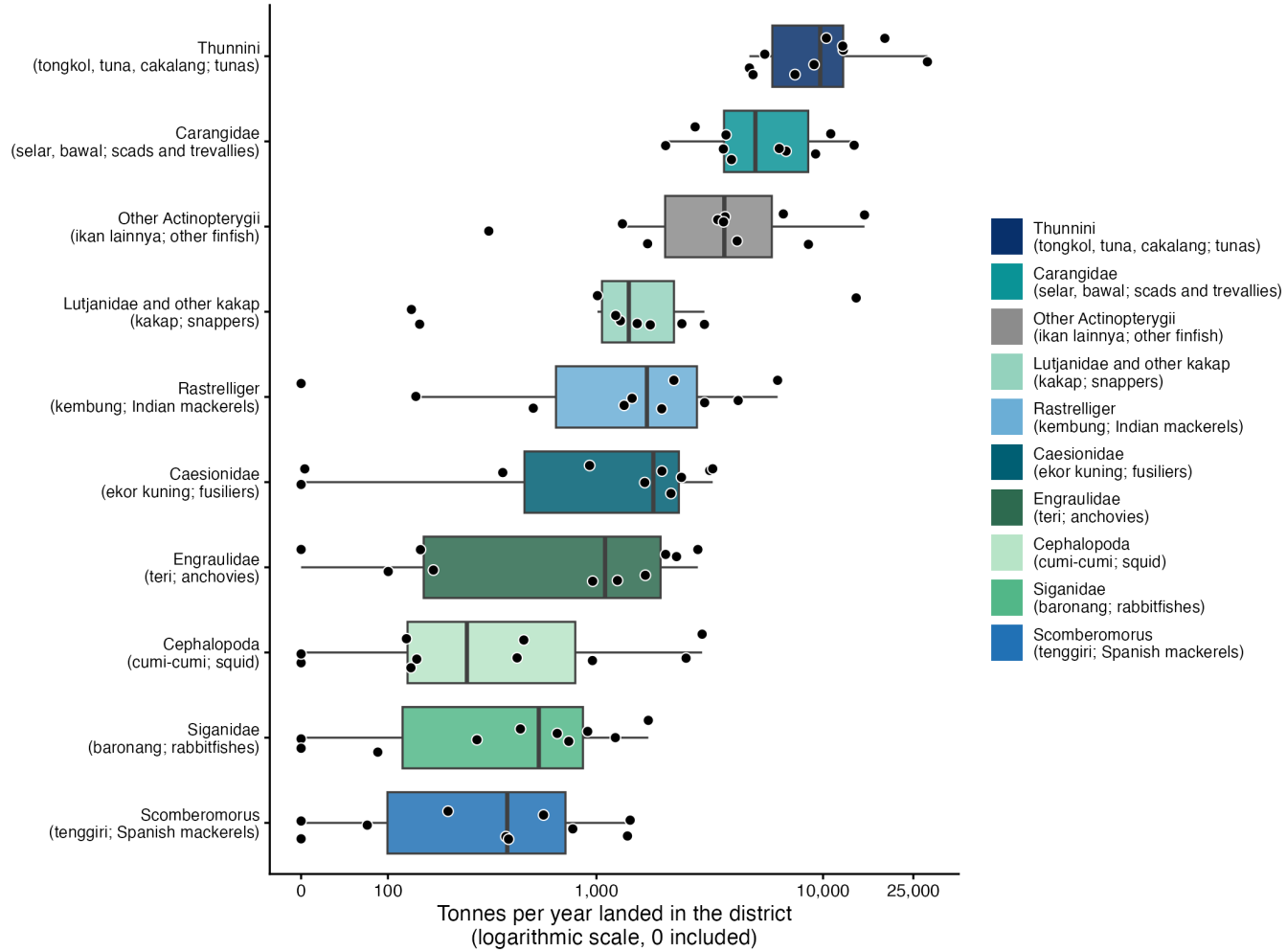


Figure 4. Marine capture of each species group across the ten districts, North Maluku, 2020. Each dot is a district; the box covers the middle half of the ten districts and the line inside it marks the median. The axis is logarithmic, with districts without landings at 0.

Findings

- The median district lands 9,735 tonnes of tuna and 5,178 tonnes of scads and trevallies, with the middle half of the districts between 6,031 and 12,265 tonnes of tuna and between 3,660 and 8,672 tonnes of scads and trevallies. The other eight groups stay below 4,000 tonnes in the median district.
- The six smaller groups are absent from one or two districts each (Pulau Morotai for four of them) and spread widely elsewhere, and in five groups one or two districts land three to ten times the median (Ternate for tuna, Halmahera Selatan for other finfish and snappers, Halmahera Barat for Indian mackerels, Halmahera Timur and Halmahera Utara for squid), so the catch of these groups is concentrated in a few districts.

2.1.3 Marine Capture Across Districts

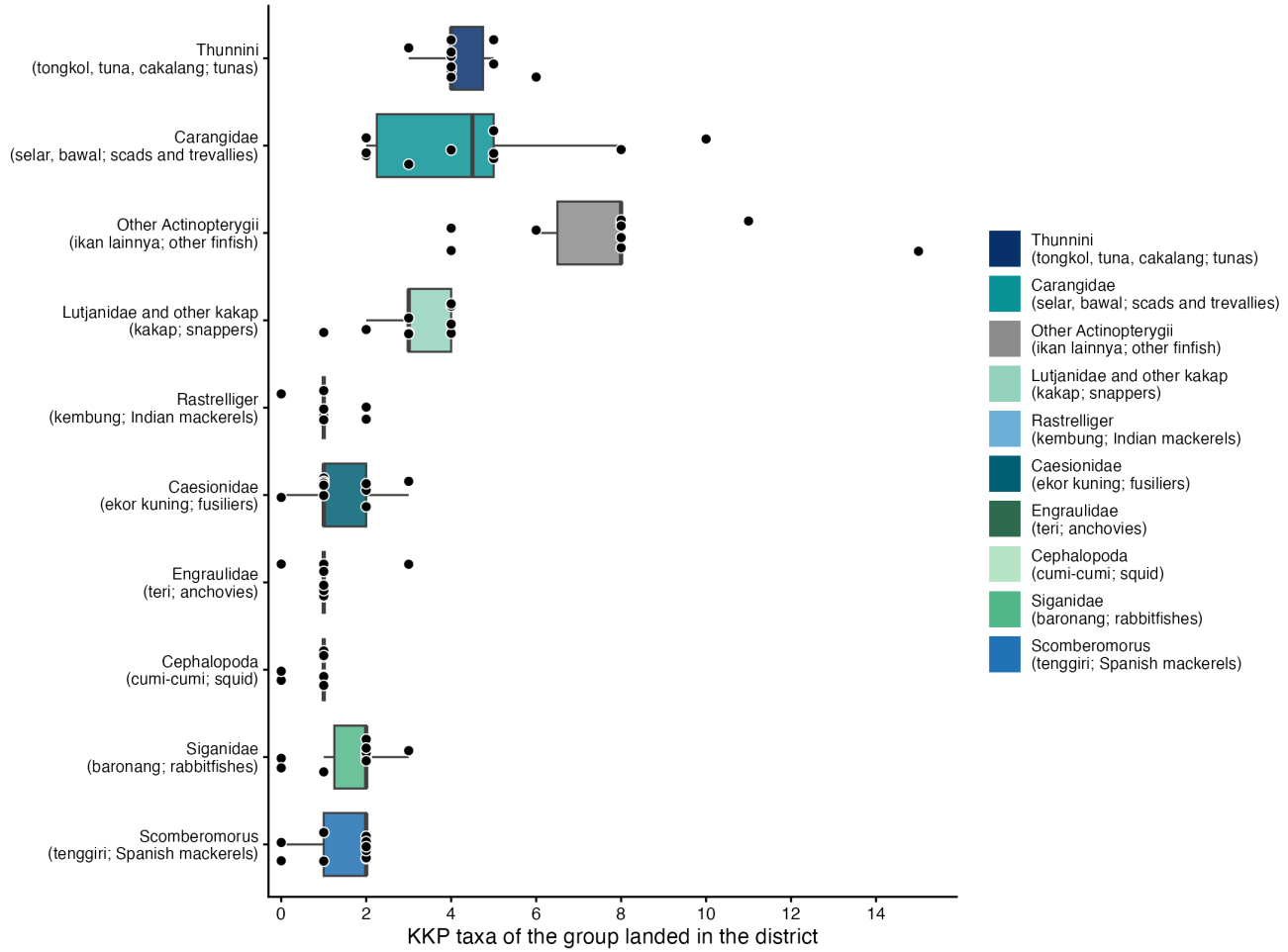


Figure 5. Number of KKP taxa of each species group landed per district, North Maluku, 2020. Each dot is a district; the box covers the middle half of the ten districts and the line inside it marks the median, and a district without landings of the group sits at 0.

Findings

- The median district lands 4 of the 7 tuna taxa, between 4 and 5 of the 14 scad and trevally taxa, 8 of the 23 other finfish taxa, 3 of the 8 snapper taxa, and one or two taxa of each of the six smaller groups, so a typical district fishes a narrow set of species within each group.
- The spread between districts is small, the middle half of the districts differing by at most three taxa within any group, but the counts are skewed upwards for the two largest groups, where one district (Ternate) lands 15 of the 23 other finfish taxa and 10 of the 14 scad and trevally taxa, about twice the median, and two districts (Pulau Morotai and Pulau Taliabu) land only 4 other finfish taxa. The variety of the catch is concentrated in one district, while the smaller groups are landed as the same one or two taxa nearly everywhere.

2.1.3 Marine Capture Across Districts

Table 2. Marine capture of each species group by district, North Maluku, 2020 (tonnes per year), with the mean, standard deviation (SD), and coefficient of variation (CV) of the ten district values. A blank cell marks a group not landed in the district.

Species Group	Halmahera Barat	Halmahera Tengah	Kepulauan Sula	Halmahera Selatan	Halmahera Utara	Halmahera Timur	Pulau Morotai	Pulau Taliabu	Ternate	Tidore Kepulauan	Mean	SD	CV
Thunnini	4,735	5,534	12,284	7,521	10,344	4,913	12,209	9,127	28,851	18,721	11,424	7,467	0.65
Carangidae	10,811	6,872	2,022	9,272	3,635	3,948	2,729	6,409	3,733	13,716	6,315	3,870	0.61
Other Actinopterygii	1,303	3,698	6,683	15,233	4,181	3,432	3,642	1,683	8,620	329	4,880	4,393	0.90
Lutjanidae and other kakap	152	2,384	1,730	13,998	2,999	1,010	1,277	1,515	1,216	137	2,642	4,086	1.55
Rastrelliger	6,305	4,225	3,005	145	2,199	1,326		1,943	523	1,438	2,111	1,967	0.93
Caesionidae	933	4	1,635	2,375	1,947	3,164		381	3,262	2,134	1,583	1,213	0.77
Engraulidae	962		1,239	2,022	2,805	2,259	100	1,645	153	178	1,136	1,021	0.90
Cephalopoda	474		442	960	2,489	2,929		128	146	136	771	1,066	1.38
Siganidae		290	1,696	914	1,212	669		754	458	86	608	558	0.92
Scomberomorus			393	1,372	211	582	405	1,410	785	72	523	522	1.00
All ten groups	**25,675**	**23,006**	**31,129**	**53,812**	**32,021**	**24,231**	**20,361**	**24,994**	**47,749**	**36,947**	**31,993**	**11,130**	**0.35**

Findings

- Tuna is landed in every district, from 4,735 tonnes in Halmahera Barat to 28,851 tonnes in Ternate, and so are scads and trevallies, from 2,022 tonnes in Kepulauan Sula to 13,716 tonnes in Tidore Kepulauan. Their landings are the most evenly spread across the province (CV 0.65 and 0.61).
- The landings of the kakap group and squid are the most uneven (CV 1.55 and 1.38) because Halmahera Selatan lands 13,998 of the 26,417 tonnes of kakap and Halmahera Timur 2,929 of the 7,705 tonnes of squid. Six groups are missing from one or two districts, and Pulau Morotai lacks four of them.

2.2.1 Household Consumption as a Share of Marine Capture

Table 3. The ten species groups that link the SUSENAS aquatic foods to the KKP taxonomy, with the SUSENAS aquatic foods of each group and their taxonomy and the KKP taxa of each group with their scientific names, North Maluku, 2020.

Species Group	SUSENAS Aquatic Food (Code) and Taxonomy	KKP Fish Group	KKP Taxonomy	Scientific Name	Basis for Grouping
Thunnini (tongkol, tuna, cakalang; tunas)	Tongkol, tuna, cakalang (18, 41)	Cakalang	*Cakalang*	*Katsuwonus pelamis*	All are tunas of the tribe Thunnini, which
	Thunnus, Euthynnus affinis, Actinopterygii	Tongkol	*Tongkol abu-abu*	*Thunnus tonggol*	SUSENAS records as tongkol, tuna, and
		Tongkol	*Tongkol banyar*	*Euthynnus lineatus*	cakalang in one aquatic food.
		Tongkol	*Tongkol komo*	*Euthynnus affinis*	
		Tongkol	*Tongkol*	Scombridae	
		Tuna	*Tuna mata besar*	*Thunnus obesus*	
		Tuna	*Tuna sirip kuning; Madidihang*	*Thunnus albacares*	
Carangidae (selar, bawal; scads and trevallies)	Selar (20, 43), bawal (31), Carangidae	Bawal	*Bawal hitam*	*Parastromateus niger*	All belong to the family Carangidae, the taxon
		Kuwe	*Kuwe sirip biru*	*Caranx melampygus*	shared with the SUSENAS aquatic food selar,
		Kuwe	*Kuwe sirip kuning*	*Caranx heberi*	and the black pomfret is also the SUSENAS
		Kuwe	*Kuwe*	*Caranx*	aquatic food bawal.
		Layang	*Layang anggur*	*Decapterus kurroides*	
		Layang	*Layang benggol*	*Decapterus russelli*	
		Layang	*Layang biru*	*Decapterus macarellus*	
		Layang	*Layang lidi; Layar deles*	*Decapterus macrosoma*	
		Layang	*Layang*	*Decapterus*	
		Selar	*Selar bentong*	*Selar crumenophthalmus*	
		Selar	*Selar hijau*	*Atule mate*	
		Selar	*Selar kuning*	*Selaroides leptolepis*	
		Selar	*Tetengkek*	*Megalaspis cordyla*	
		Sunglir	*Sunglir*	*Elagatis bipinnulata*	
Other Actinopterygii (ikan lainnya; other finfish)	Ikan lainnya (33, 48)	Bayeman	*Kakaktua putih*	*Cheilinus chlorourus*	Marine finfish from 12 families with no SUSENAS
	Actinopterygii	Biji nangka	*Biji nangka karang*	*Parupeneus indicus*	aquatic food of their own, which SUSENAS

2.2.1 Household Consumption as a Share of Marine Capture

Table 3. The ten species groups that link the SUSENAS aquatic foods to the KKP taxonomy, with the SUSENAS aquatic foods of each group and their taxonomy and the KKP taxa of each group with their scientific names, North Maluku, 2020. (continued)

Species Group	SUSENAS Aquatic Food (Code) and Taxonomy	KKP Fish Group	KKP Taxonomy	Scientific Name	Basis for Grouping
Other Actinopterygii (continued)		Biji nangka	*Kuniran; Biji nangka*	*Upeneus moluccensis*	records only as other fresh or other preserved
		Julung-julung	*Cendro*	*Tylosurus crocodilus*	fish.
		Julung-julung	*Julung-julung*	*Hemiramphus far*	
		Kakatua	*Kakatua biru*	*Scarus ghobban*	
		Kakatua	*Kakatua*	Scarinae	
		Kerapu	*Kerapu batu*	*Scorpaenopsis oxycephala*	
		Kerapu	*Kerapu bebek; Kerapu tikus*	*Chromileptes altivelis*	
		Kerapu	*Kerapu karang*	*Cephalopholis boenak*	
		Kerapu	*Kerapu lumpur*	*Epinephelus coioides*	
		Kerapu	*Kerapu merah bintik biru*	*Plectropomus leopardus*	
		Kerapu	*Kerapu merah*	*Cephalopholis sonnerati*	
		Kerapu	*Kerapu*	Epinephelidae	
		Kurisi	*Kerisi*	*Nemipterus hexodon*	
		Kurisi	*Kurisi bali*	*Nemipterus balinensis*	
		Kurisi	*Kurisi*	*Nemipterus hexodon*	
		Layur	*Cendro*	*Trichiurus auriga*	
		Layur	*Layur*	*Trichiurus lepturus*	
		Lemuru	*Tembang*	*Sardinella fimbriata*	
		Lencam	*Lencam*	*Lethrinus nebulosus*	
		Swanggi	*Swanggi*	*Priacanthus tayenus*	
		Terbang	*Terbang; Torani*	*Hirundichthys oxycephalus*	
Lutjanidae and other kakap (kakap; snappers)	Kakap (28)	Kaci-kaci	*Kakap putih*	*Plectorhinchus gibbosus*	Snappers of the family Lutjanidae, the taxon
	Lutjanus	Kakap	*Kakap batu*	*Lobotes surinamensis*	shared with the SUSENAS aquatic food kakap,

2.2.1 Household Consumption as a Share of Marine Capture

Table 3. The ten species groups that link the SUSENAS aquatic foods to the KKP taxonomy, with the SUSENAS aquatic foods of each group and their taxonomy and the KKP taxa of each group with their scientific names, North Maluku, 2020. (continued)

Species Group	SUSENAS Aquatic Food (Code) and Taxonomy	KKP Fish Group	KKP Taxonomy	Scientific Name	Basis for Grouping
Lutjanidae and other kakap (continued)		Kakap	*Kakap merah utara*	*Lutjanus bohar*	together with the three fish of other families
		Kakap	*Kakap merah*	*Lutjanus malabaricus*	that are landed and sold as kakap.
		Kakap	*Kakap putih; Baramundi*	*Lates calcarifer*	
		Kakap	*Kakap*	*Lutjanus*	
		Kakap	*Kurisi bali*	*Pristipomoides typus*	
		Kakap	*Pinjalo*	*Pinjalo pinjalo*	
Rastrelliger (kembung; Indian mackerels)	Kembung (21, 39)	Kembung	*Kembung lelaki*	*Rastrelliger kanagurta*	Both are Indian mackerels of the genus
	Rastrelliger	Kembung	*Kembung*	*Rastrelliger*	Rastrelliger, the taxon shared with the SUSENAS aquatic food kembung.
Caesionidae (ekor kuning; fusiliers)	Ekor kuning (17)	Ekor kuning	*Ekor kuning*	*Caesio cuning*	All are fusiliers of the family Caesionidae,
	Caesio cuning	Ekor kuning	*Lolosi papan*	*Caesio xanthonota*	the taxon shared with the SUSENAS aquatic food
		Pisang-pisang	*Pisang-pisang*	*Pterocaesio pisang*	ekor kuning.
Engraulidae (teri; anchovies)	Teri (22, 42)	Teri	*Teri bulu ayam*	*Thryssa mystax*	All are anchovies of the family Engraulidae,
	Stolephorus	Teri	*Teri merek*	*Encrasicholina heteroloba*	the taxon shared with the SUSENAS aquatic food
		Teri	*Teri nasi*	*Stolephorus tri*	teri.
Cephalopoda (cumi-cumi; squid)	Cumi-cumi (35, 50) Cephalopoda	Cumi-cumi	*Cumi-cumi*	Loliginidae	Squid are the only cephalopods in the capture statistics, and SUSENAS records squid, cuttlefish, and octopus in one aquatic food.
Siganidae (baronang; rabbitfishes)	Baronang (29)	Baronang	*Baronang angin*	*Siganus javus*	All are rabbitfishes of the genus Siganus,
	Siganus	Baronang	*Baronang hitam*	*Siganus fuscescens*	the taxon shared with the SUSENAS aquatic food
		Baronang	*Baronang kunyit*	*Siganus puellus*	baronang.
		Baronang	*Baronang*	*Siganus guttatus*	
Scomberomorus (tenggiri; Spanish mackerels)	Tenggiri (19, 40)	Tenggiri	*Tenggiri batang; Tenggiri India*	*Scomberomorus commerson*	All are Spanish mackerels of the genus
	Scomberomorus	Tenggiri	*Tenggiri papan*	*Scomberomorus guttatus*	Scomberomorus, the taxon shared with the
		Tenggiri	*Tenggiri*	*Scomberomorus*	SUSENAS aquatic food tenggiri.

Findings

- The ten species groups link the 68 KKP taxa to 18 SUSENAS aquatic foods through a taxon that both sources share, from a single species (Caesio cuning for ekor kuning) to a whole class (Actinopterygii for ikan lainnya).
- Eight groups are matched at genus, family, or tribe level. Other Actinopterygii pools 23 taxa from 12 families behind one SUSENAS aquatic food, ikan lainnya, and Cephalopoda pools squid behind an aquatic food that also covers cuttlefish and octopus, so these two class-level groups are the loosest links.

2.2.1 Household Consumption as a Share of Marine Capture

Table 4. Marine capture and household consumption by district, North Maluku, 2020, with the landings per person in kilograms per year (population from the survey weights), consumption as a share of the district's marine capture, and the numbers of KKP taxa, species groups, and SUSENAS aquatic foods landed or reported.

	Marine Capture					Household Consumption				
District	**Tonnes**	**Share of Province (%)**	**Per Person (kg)**	**KKP Taxa Landed (of 68)**	**Species Groups Landed (of 10)**	**Tonnes**	**Share of Province (%)**	**Share of Marine Capture (%)**	**SUSENAS Aquatic Foods Reported (of 35)**	**Species Groups Consumed (of 10)**
Halmahera Barat	25,675	8.0	214.5	19	8	3,884	8.4	15.1	29	10
Halmahera Tengah	23,006	7.2	405.4	21	7	2,247	4.9	9.8	28	10
Kepulauan Sula	31,129	9.7	295.2	27	10	4,442	9.6	14.3	20	10
Halmahera Selatan	53,812	16.8	226.3	37	10	10,009	21.7	18.6	24	10
Halmahera Utara	32,021	10.0	162.9	30	10	6,385	13.8	19.9	24	9
Halmahera Timur	24,231	7.6	250.3	27	10	3,429	7.4	14.2	27	10
Pulau Morotai	20,361	6.4	297.0	16	6	1,996	4.3	9.8	24	10
Pulau Taliabu	24,994	7.8	468.8	19	10	2,435	5.3	9.7	27	10
Ternate	47,749	14.9	202.4	47	10	7,856	17.0	16.5	20	10
Tidore Kepulauan	36,947	11.5	361.7	31	10	3,541	7.7	9.6	20	10
North Maluku	**319,925**	**100.0**	**251.3**	**68**	**10**	**46,226**	**100.0**	**14.4**	**32**	**10**

Findings

- Halmahera Selatan lands 53,812 tonnes (16.8% of the province) and consumes 10,009 tonnes (21.7%), and Ternate lands 47,749 tonnes and consumes 7,856 tonnes, so the two largest fisheries are also the two largest consumers, together 32% of the catch and 39% of household consumption.
- Households consume between 9.6% (Tidore Kepulauan) and 19.9% (Halmahera Utara) of what their district lands, 14.4% in the province, which means that more than four fifths of the landings are not consumed by the households of the district that lands them.
- Ternate lands 47 of the 68 KKP taxa, yet its households report only 20 of the 35 SUSENAS aquatic foods, the fewest together with Kepulauan Sula and Tidore Kepulauan, so a diverse catch does not bring a diverse fish consumption.

2.2.1 Household Consumption as a Share of Marine Capture

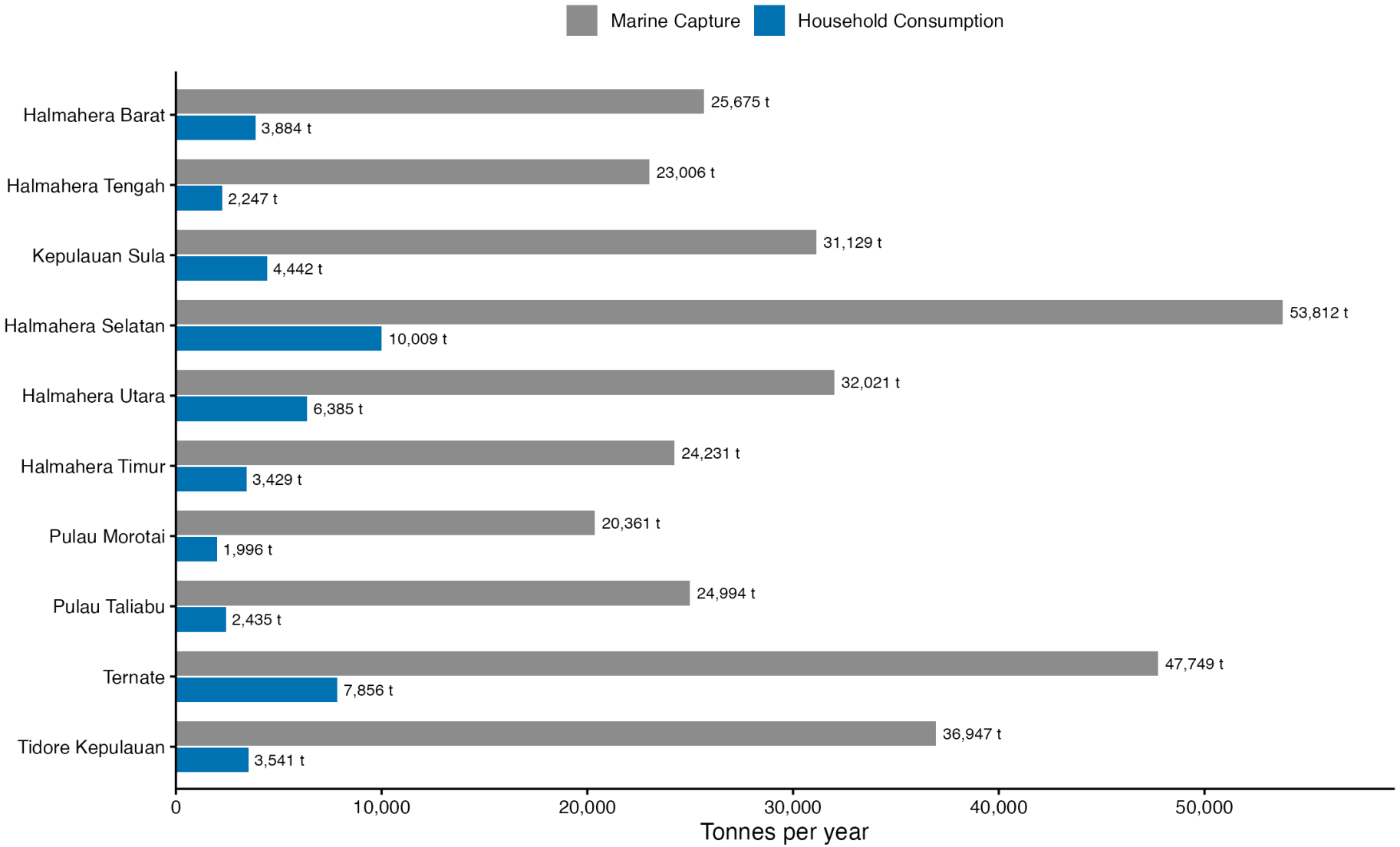


Figure 6. Marine capture and household fish consumption by district, North Maluku, 2020 (tonnes per year).

Findings

- Landings exceed household consumption in every district, five times over in Halmahera Utara and ten times over in Tidore Kepulauan, and the two largest fisheries, Halmahera Selatan and Ternate, are also the two largest consumers.
- In Halmahera Tengah, Pulau Morotai, Pulau Taliabu, and Tidore Kepulauan household consumption is under 10% of the local landings, so most of the fish landed in these districts goes elsewhere.

2.2.1 Household Consumption as a Share of Marine Capture

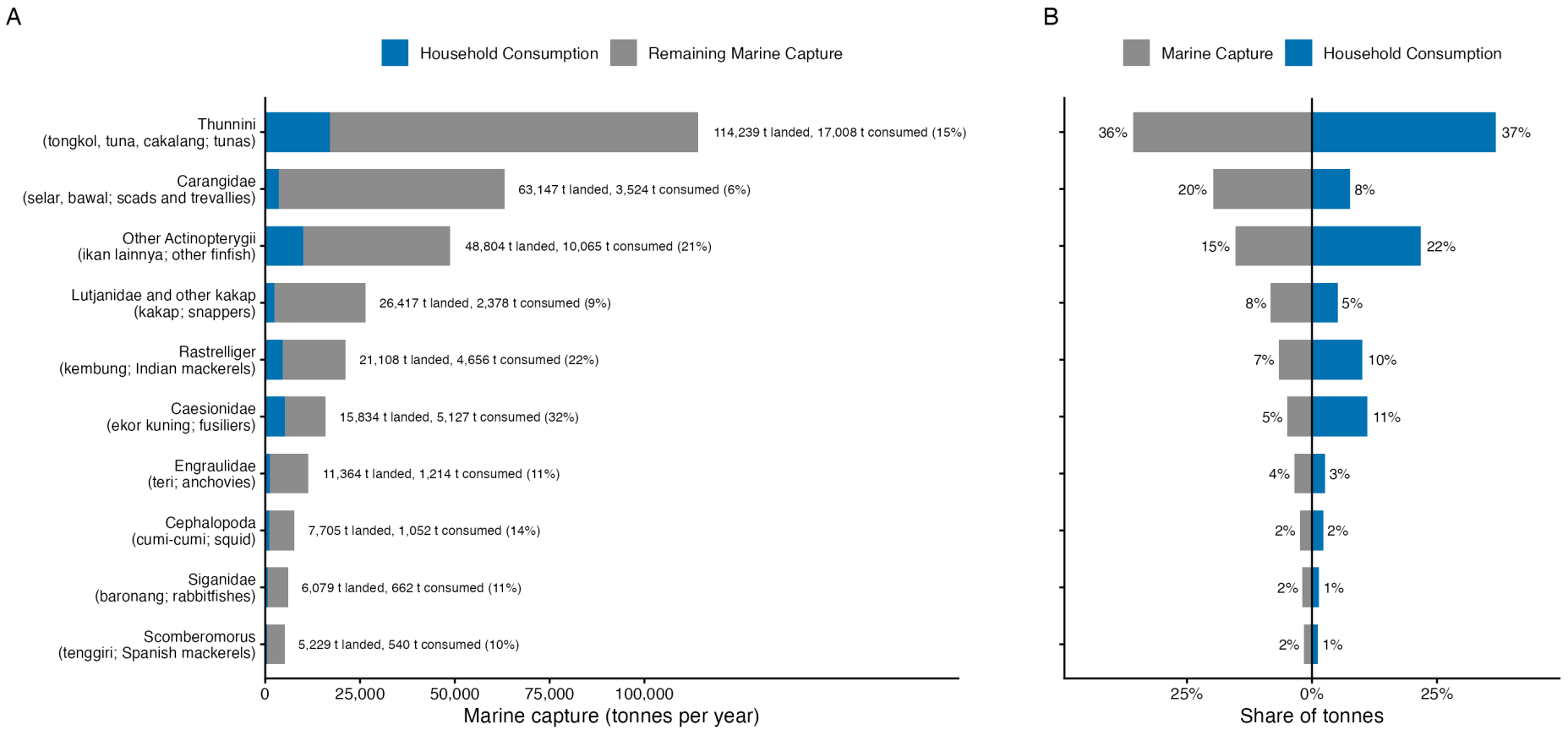


Figure 7. Marine capture and household fish consumption by species group, North Maluku, 2020. A, tonnes landed, with the part consumed by households in blue and the label giving the tonnes landed, the tonnes consumed, and the consumed share. B, species composition of marine capture and of household fish consumption as shares of the tonnes of the ten groups.

Findings

- Households consume 14% of the tonnes landed of the ten groups (panel A), from 6% of Carangidae, the second largest group landed, to 32% of Caesionidae (fusiliers), and tuna keeps nearly the same share on both sides, 36% of the catch and 37% of consumption.
- In panel B, other finfish are 15% of the catch but 22% of household consumption, and fusiliers and Indian mackerels also make up a larger share of consumption than of the catch, whereas scads and trevallies are 20% of the catch but only 8% of consumption and the kakap group also falls. Households therefore consume relatively more reef fish and small mackerels than the fishery lands.

2.2.3 Household Consumption by District and by Species Group

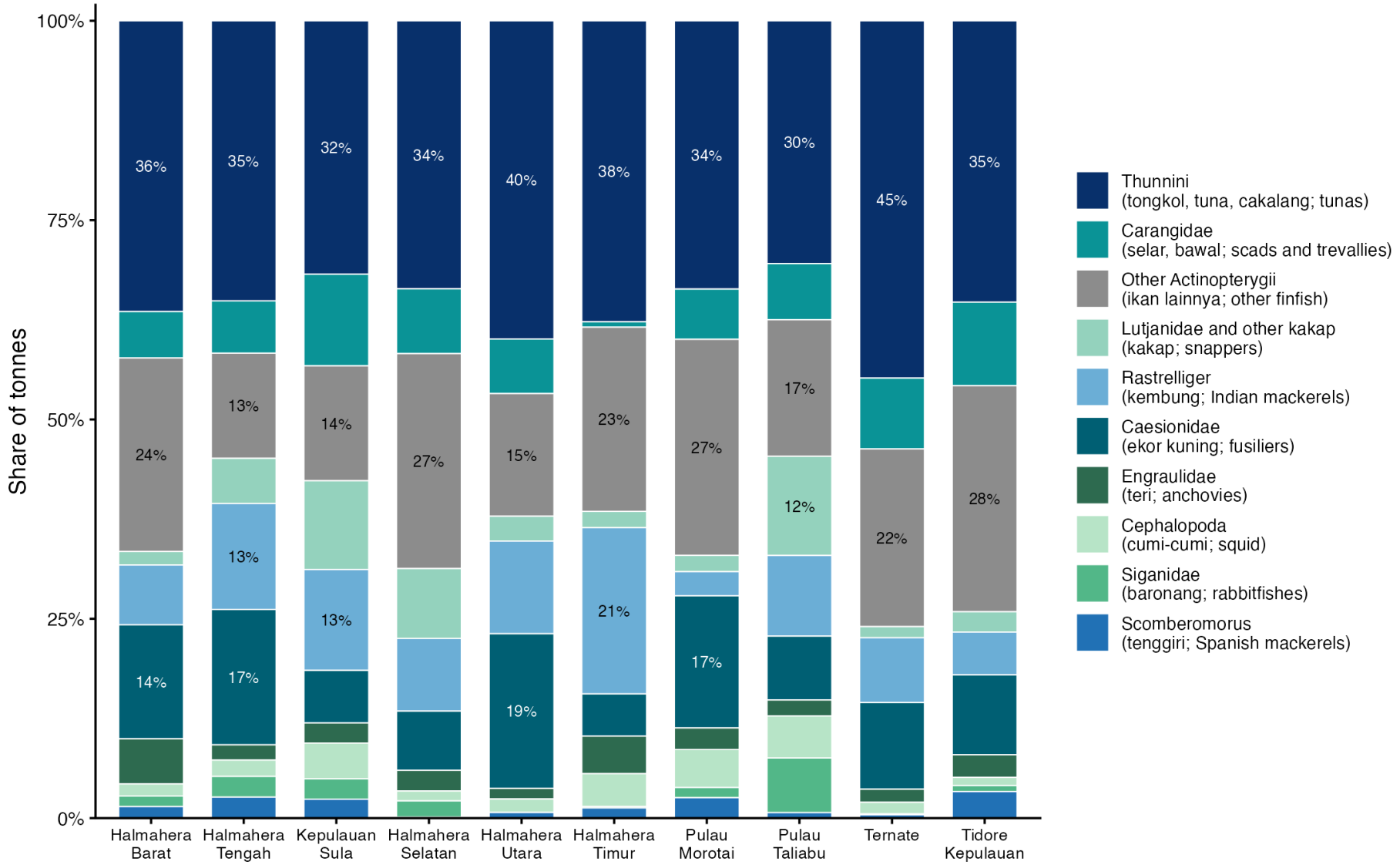


Figure 8. Species composition of household fish consumption by district, North Maluku, 2020. Shares of at least 12% are labelled.

Findings

- Tuna is the most consumed group in every district, 30% to 45% of household fish consumption, whatever the local catch. In Pulau Morotai tuna is 60% of the landings but 34% of consumption. Other finfish come second in eight of the ten districts.
- The leading group holds 30% to 45% of consumption in every district, against 20% to 60% of the landings, so household consumption is more alike across the province than the catch.

2.2.3 Household Consumption by District and by Species Group

Table 5. Household consumption of each species group by district, North Maluku, 2020 (tonnes per year), with the mean, standard deviation (SD), and coefficient of variation (CV) of the ten district values. A blank cell marks a group not consumed in the district.

Species Group	Halmahera Barat	Halmahera Tengah	Kepulauan Sula	Halmahera Selatan	Halmahera Utara	Halmahera Timur	Pulau Morotai	Pulau Taliabu	Ternate	Tidore Kepulauan	Mean	SD	CV
Thunnini	1,416	789	1,412	3,365	2,548	1,295	672	741	3,520	1,249	1,701	1,061	0.62
Carangidae	226	147	510	814	437	23	126	171	698	371	352	261	0.74
Other Actinopterygii	943	296	640	2,698	982	792	541	417	1,751	1,004	1,006	720	0.72
Lutjanidae and other kakap	66	128	495	878	199	70	41	303	109	90	238	264	1.11
Rastrelliger	291	299	562	912	742	715	60	246	639	190	466	284	0.61
Caesionidae	555	382	293	744	1,238	181	331	195	853	355	513	339	0.66
Engraulidae	220	43	112	260	85	162	54	49	129	101	121	73	0.60
Cephalopoda	59	46	199	124	108	141	95	128	114	36	105	49	0.47
Siganidae	51	58	114	204		6	26	167	10	26	66	71	1.08
Scomberomorus	56	59	106	12	45	44	51	17	32	118	54	34	0.63
All ten groups	**3,884**	**2,247**	**4,442**	**10,009**	**6,385**	**3,429**	**1,996**	**2,435**	**7,856**	**3,541**	**4,623**	**2,647**	**0.57**

Findings

- Household consumption is spread more evenly across the districts than the catch (CV below 0.75 for eight of the ten groups, against 0.61 to 1.55 for the landings), and every group is consumed in all ten districts except Siganidae (rabbitfishes), which has no record in Halmahera Utara.
- Halmahera Selatan consumes 878 of the 2,378 tonnes of the kakap group and 204 of the 662 tonnes of Siganidae, more than a third of each, so these two groups are the least evenly consumed (CV 1.11 and 1.08). The same district is the largest consumer of six of the ten groups, as it is the largest fishery.

2.2.3 Household Consumption by District and by Species Group

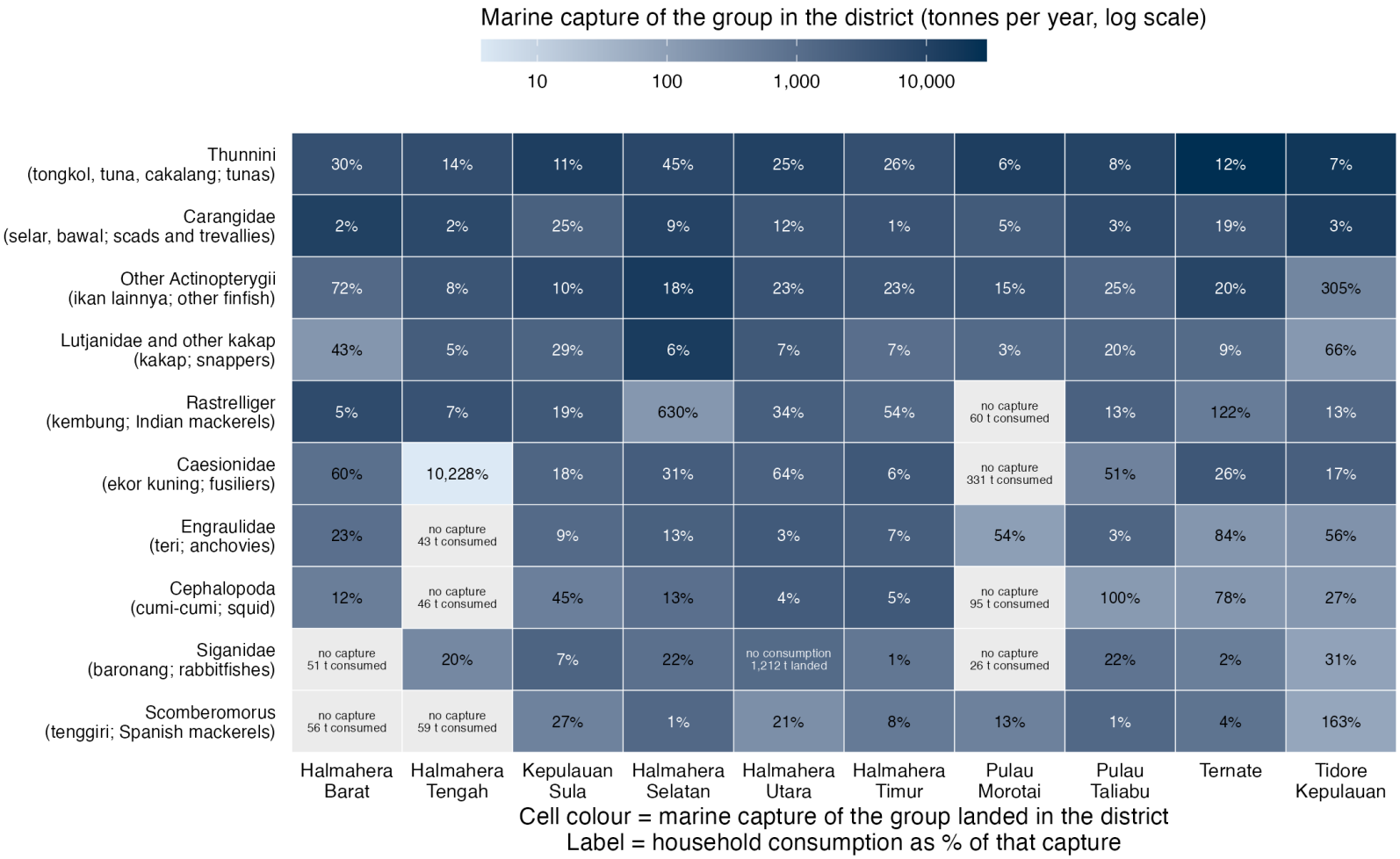


Figure 9. Household fish consumption as a share of the marine capture of the same species group in the same district, North Maluku, 2020. The colour of a cell is the volume of the group landed in the district on a logarithmic scale, the label is the consumption as a share of that volume, grey cells mark groups not landed in the district and give the tonnes consumed there, and the one cell with landings but no consumption gives the tonnes landed.

Findings

- Household consumption of a group is usually a small part of the landings of the same group in the same district, between 5% and 30% in most districts for most groups, and for scads and trevallies it never passes 25% of the local landings.
- In five cases household consumption of a group exceeds the landings recorded for the same group in the same district, for example fusiliers in Halmahera Tengah (10,228%) and Indian mackerels in Halmahera Selatan (630%), so for these groups the district's recorded landings do not account for what its households consume.

2.2.4 Household Consumption of Aquatic Foods Outside Marine Capture

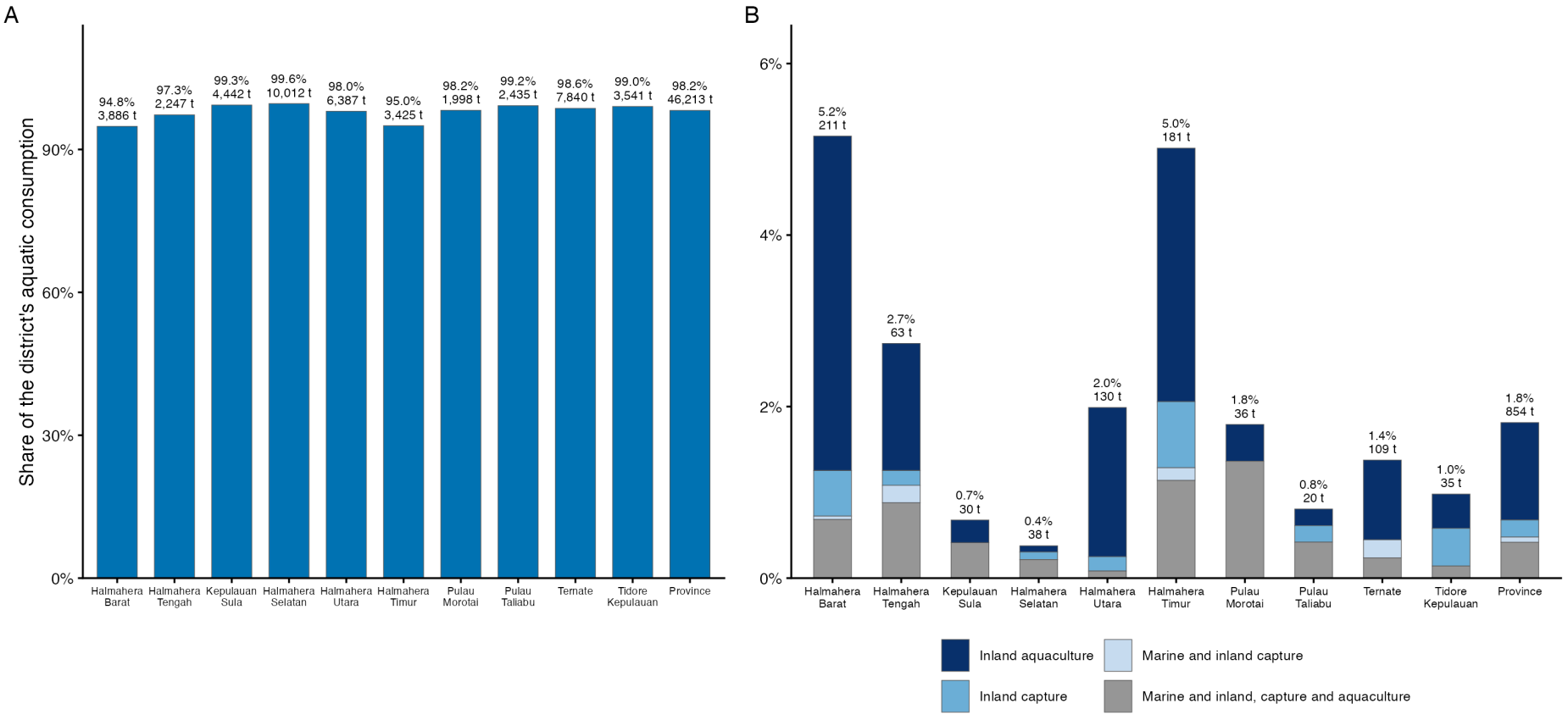


Figure 10. Household consumption of SUSENAS aquatic foods by habitat and production method and by district, North Maluku, 2020. A, the share of the district's aquatic consumption classed as marine capture. B, the shares of the four other classes, stacked, on their own scale. The label above a bar gives its total in percent and in tonnes per year.

Findings

- Fish from marine capture makes up 98% of household aquatic consumption in the province and at least 95% in every district, and Halmahera Selatan, Kepulauan Sula, Pulau Taliabu, and Tidore Kepulauan reach 99% or more.
- Farmed freshwater fish is the only other class of note. It reaches 3.9% of aquatic consumption in Halmahera Barat and 3.0% in Halmahera Timur and stays under 2% in the other eight districts.

2.2.4 Household Consumption of Aquatic Foods Outside Marine Capture

Table 6. SUSENAS aquatic foods consumed by households but absent from the marine capture statistics, by habitat and production method, North Maluku, 2020.

Habitat and Production Method	SUSENAS Aquatic Food (Code)	Tonnes per Year	Share of Aquatic Consumption	Districts With Records	Spread Across Districts (t)			
					Min	Max	Mean	SD
Marine capture	Ikan dalam kaleng (47), canned fish	15.5	0.03%	8	0.0	4.6	1.6	1.6
Inland aquaculture	Mujair (25), Mozambique tilapia	267.6	0.57%	8	0.0	90.2	26.8	37.8
	Mas, nila (26), common carp and Nile tilapia	121.3	0.26%	6	0.0	57.2	12.1	19.9
	Gurame (32), giant gourami	52.5	0.11%	7	0.0	23.7	5.3	7.2
	Lele (27), catfish	40.1	0.09%	6	0.0	12.9	4.0	5.1
	Patin (30), pangasius	26.7	0.06%	4	0.0	17.2	2.7	5.5
	Bandeng (23), milkfish	26.3	0.06%	7	0.0	10.3	2.6	3.9
	Sepat diawetkan (44), preserved gourami	0.0	0.00%	0	0.0	0.0	0.0	0.0
	Bandeng diawetkan (45), preserved milkfish	0.0	0.00%	0	0.0	0.0	0.0	0.0
Inland capture	Gabus (24), snakehead	93.3	0.20%	7	0.0	27.8	9.3	9.7
	Gabus diawetkan (46), preserved snakehead	0.3	0.00%	1	0.0	0.3	0.0	0.1
Marine and inland, capture and aquaculture	Udang, lobster (34), shrimp and lobster	86.0	0.18%	10	1.9	18.8	8.6	5.7
	Ketam, kepiting, rajungan (36), crabs	70.1	0.15%	7	0.0	25.8	7.0	8.7
	Kerang, siput, bekicot, remis (37), shellfish and snails	20.7	0.04%	5	0.0	6.3	2.1	2.5
	Other fresh aquatic animals (38)	18.5	0.04%	3	0.0	15.2	1.9	4.8
	Other preserved aquatic animals (51)	1.7	0.00%	2	0.0	1.1	0.2	0.4
	Udang diawetkan (49), dried shrimp (ebi, rebon)	0.0	0.00%	0	0.0	0.0	0.0	0.0
Total, 17 foods		**840.8**	**1.79%**					

Findings

- The 17 SUSENAS aquatic foods with no counterpart in the marine capture statistics add up to 840.8 tonnes a year, 1.8% of aquatic consumption, so the comparison with the capture statistics covers almost all household aquatic consumption.
- Most of this small remainder, 534.6 tonnes, is farmed freshwater fish such as mujair, mas and nila, and gurame. Only udang and lobster (shrimp and lobster) is consumed in all ten districts, and three preserved aquatic foods have no record at all.